

Extensions of PerfectGroup(9720,2) via $Z\phi$ -classes

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Solvable quotient groups of orders $2 \leq n \leq 64$

Here $|S| = 2n$ and $|E| = 9720n$. GAP runtimes are reported in seconds.

Let $P = \text{PerfectGroup}(9720, 2)$. In this computation one has

$$Z(P) \simeq C_2, \quad \text{Out}(P) \simeq C_6, \quad P/Z(P) \simeq \text{PerfectGroup}(4860, 2).$$

Since $\text{Aut}(C_2) = 1$, there is no marking ambiguity and no non-trivial central action to record. The auxiliary group S has order $2n$ and contains a distinguished central involution z identified with $Z(P)$. For each orbit of such central involutions under $\text{Aut}(S)$, the quotient

$$H = S/\langle z \rangle$$

is tested for all couplings

$$\phi : H \longrightarrow \text{Out}(P) \simeq C_6.$$

The computation is not restricted to surjective couplings. The possible images are exactly

$$1, \quad C_2, \quad C_3, \quad C_6,$$

and they are displayed below as the columns *Central*, C_2 , C_3 and C_6 .

How the couplings are classified. Fix a central involution $z \in Z(S)$ and put $H = S/\langle z \rangle$. A coupling with image $U \leq C_6$ is determined by a normal subgroup $K \trianglelefteq H$ with $H/K \simeq U$, together with an identification $H/K \rightarrow U$. Two such identifications may define equivalent $Z\phi$ -data if they differ by an automorphism induced from the stabilizer of z in $\text{Aut}(S)$. Equivalently, for each kernel K , one counts the corresponding double-coset quotient inside $\text{Aut}(U)$, as in the GrpConst-style classification of couplings. Since C_6 is abelian, the normalizer action of $N_{C_6}(U)$ on U is trivial; thus no additional conjugation inside $\text{Out}(P)$ identifies the different embeddings. This explains why the non-surjective couplings with images C_2 and C_3 must be counted separately from the surjective image C_6 .

Table of $Z\phi$ -classes

The following tables use almost the full available text width. The columns *Cumulative* and *Cum. time* are global cumulative sums from $n = 2$ onward.

Values for $2 \leq n \leq 17$.

n	$ S $	$ E $	Central	C_2	C_3	C_6	$\#Z\phi$	Cumulative	Time (s)	Cum. time (s)
2	4	19 440	2	2	0	0	4	4	0.337	0.337
3	6	29 160	1	0	1	0	2	6	0.085	0.422
4	8	38 880	6	8	0	0	14	20	0.558	0.980
5	10	48 600	1	0	0	0	1	21	0.058	1.038
6	12	58 320	4	4	2	2	12	33	6.965	8.003
7	14	68 040	1	0	0	0	1	34	0.023	8.026
8	16	77 760	21	42	0	0	63	97	0.745	8.771
9	18	87 480	2	0	2	0	4	101	0.064	8.835
10	20	97 200	4	4	0	0	8	109	0.439	9.274
11	22	106 920	1	0	0	0	1	110	0.021	9.295
12	24	116 640	16	24	8	8	56	166	1.303	10.598
13	26	126 360	1	0	0	0	1	167	0.018	10.616
14	28	136 080	4	4	0	0	8	175	0.199	10.815
15	30	145 800	1	0	1	0	2	177	0.028	10.843
16	32	155 520	95	286	0	0	381	558	6.932	17.775
17	34	165 240	1	0	0	0	1	559	0.020	17.795

Values for $18 \leq n \leq 33$.

n	$ S $	$ E $	Central	C_2	C_3	C_6	$\#Z\phi$	Cumulative	Time (s)	Cum. time (s)
18	36	174 960	10	10	6	6	32	591	0.954	18.749
19	38	184 680	1	0	0	0	1	592	0.024	18.773
20	40	194 400	16	26	0	0	42	634	1.251	20.024
21	42	204 120	2	0	3	0	5	639	0.150	20.174
22	44	213 840	4	4	0	0	8	647	0.227	20.401
23	46	223 560	1	0	0	0	1	648	0.029	20.430
24	48	233 280	72	166	26	46	310	958	9.020	29.450
25	50	243 000	2	0	0	0	2	960	0.039	29.489
26	52	252 720	4	4	0	0	8	968	0.310	29.799
27	54	262 440	5	0	8	0	13	981	0.447	30.246
28	56	272 160	14	24	0	0	38	1 019	1.518	31.764
29	58	281 880	1	0	0	0	1	1 020	0.021	31.785
30	60	291 600	8	8	4	4	24	1 044	1.744	33.529
31	62	301 320	1	0	0	0	1	1 045	0.133	33.662
32	64	311 040	649	3 026	0	0	3 675	4 720	233.986	267.648
33	66	320 760	1	0	1	0	2	4 722	0.030	267.678

Values for $34 \leq n \leq 49$.

n	$ S $	$ E $	Central	C_2	C_3	C_6	$\#Z\phi$	Cumulative	Time (s)	Cum. time (s)
34	68	330 480	4	4	0	0	8	4 730	0.307	267.985
35	70	340 200	1	0	0	0	1	4 731	0.026	268.011
36	72	349 920	48	80	26	32	186	4 917	7.855	275.866
37	74	359 640	1	0	0	0	1	4 918	0.017	275.883
38	76	369 360	4	4	0	0	8	4 926	0.199	276.082
39	78	379 080	2	0	3	0	5	4 931	0.133	276.215
40	80	388 800	73	176	0	0	249	5 180	15.513	291.728
41	82	398 520	1	0	0	0	1	5 181	0.018	291.746
42	84	408 240	12	12	12	12	48	5 229	2.016	293.762
43	86	417 960	1	0	0	0	1	5 230	0.049	293.811
44	88	427 680	14	24	0	0	38	5 268	1.637	295.448
45	90	437 400	2	0	2	0	4	5 272	0.095	295.543
46	92	447 120	4	4	0	0	8	5 280	0.224	295.767
47	94	456 840	1	0	0	0	1	5 281	0.018	295.785
48	96	466 560	422	1 574	116	306	2 418	7 699	107.288	403.073
49	98	476 280	2	0	0	0	2	7 701	0.042	403.115

Values for $50 \leq n \leq 64$.

n	$ S $	$ E $	Central	C_2	C_3	C_6	$\#Z\phi$	Cumulative	Time (s)	Cum. time (s)
50	100	486 000	10	10	0	0	20	7 721	1.047	404.162
51	102	495 720	1	0	1	0	2	7 723	0.058	404.220
52	104	505 440	16	26	0	0	42	7 765	1.525	405.745
53	106	515 160	1	0	0	0	1	7 766	0.019	405.764
54	108	524 880	30	30	30	30	120	7 886	6.190	411.954
55	110	534 600	2	0	0	0	2	7 888	0.159	412.113
56	112	544 320	64	158	0	0	222	8 110	9.735	421.848
57	114	554 040	2	0	3	0	5	8 115	0.183	422.031
58	116	563 760	4	4	0	0	8	8 123	0.297	422.328
59	118	573 480	1	0	0	0	1	8 124	0.098	422.426
60	120	583 200	46	84	18	26	174	8 298	15.909	438.335
61	122	592 920	1	0	0	0	1	8 299	0.027	438.362
62	124	602 640	4	4	0	0	8	8 307	0.264	438.626
63	126	612 360	4	0	7	0	11	8 318	0.617	439.243
64	128	622 080	8 308	72 010	0	0	80 318	88 636	7404.015	7843.258
Range		Central	C_2	C_3	C_6	$\#Z\phi$	Cumulative	Time (s)	Cum. time (s)	
$2 \leq n \leq 64$		10 038	77 846	280	472	88 636	88 636	7843.258	7843.258	

Largest contributors to the total runtime

The total GAP runtime for $2 \leq n \leq 64$ is

$$7843.258 \text{ s} = 130.721 \text{ min} \simeq 2 \text{ h } 10 \text{ min } 43 \text{ s.}$$

The following values of n dominate the computation. Times are shown both in seconds and in minutes.

n	Time (s)	Time (min)	$\#Z\phi$	Runtime share	Class share
64	7404.015	123.400	80 318	94.40%	90.62%
32	233.986	3.900	3 675	2.98%	4.15%
48	107.288	1.788	2 418	1.37%	2.73%
60	15.909	0.265	174	0.20%	0.20%
40	15.513	0.259	249	0.20%	0.28%
56	9.735	0.162	222	0.12%	0.25%
24	9.020	0.150	310	0.12%	0.35%
36	7.855	0.131	186	0.10%	0.21%

The case $n = 64$ alone accounts for about 94.40% of the runtime and 90.62% of all computed $Z\phi$ -classes. This is consistent with the general behaviour observed throughout the project: powers of 2 produce the largest number of auxiliary groups S and the largest number of central involution orbits, hence they dominate both the enumeration and the runtime.